Algebra I: Complete Solutions

Assignments 1–3

Self-contained solution manual

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How these solutions are written

Every problem is restated. The yellow box identifies the definitions and the proof plan. The green box carries out every step. A relation is never called an equivalence relation without checking reflexivity, symmetry, and transitivity; an operation on equivalence classes is never used before proving well-definedness; and a group is never asserted without checking closure, associativity, identity, and inverses.

Throughout, $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, and $\mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$.

Part I

Assignment 1: Mappings and Relations

Chapter 1

Assignment 1 solutions

1.1 Question 1: counting injections

Question

Let A be an n-element set and B an m-element set. Find the number of injections $A \to B$ when (i) $n = m$, (ii) $n > m$, and (iii) $n < m$.

Approach and theory

An injection sends distinct elements of A to distinct elements of B. Choose an ordering $a_1, \dots, a_n$ of A and count the available choices successively. This is the multiplication principle.

Step-by-step solution

For a_1 there are m choices. After its image is chosen, a_2 has $m-1$ choices, and so on. If $n \le m$, the number is

$$m(m-1)\cdots(m-n+1) = \frac{m!}{(m-n)!}.$$

Thus:

(i) If $n = m$, the answer is $m! = n!$.

(ii) If $n > m$, the answer is 0: by the pigeonhole principle, two of the n inputs would have to share one of only m outputs.

(iii) If $n < m$, the answer is $m!/(m-n)!$.

1.2 Question 2: a bijection $\mathbb{N}^2 \to \mathbb{N}$

Question

Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $f(a, b) = 2^{a-1}(2b-1)$. Prove that f is a bijection and find f^{-1}.

Approach and theory

Every positive integer has a unique power of 2 dividing it: it can be written uniquely as $2^k q$ where $k \geq 0$ and q is positive and odd. This is the existence and uniqueness of prime factorization, applied only to the prime 2.

Step-by-step solution

Injective. Suppose

$$2^{a-1}(2b-1) = 2^{c-1}(2d-1).$$

Both $2b-1$ and $2d-1$ are odd. The exponent of 2 in the prime factorization of the left side is $a-1$, and on the right it is $c-1$. Uniqueness gives $a-1 = c-1$, so $a = c$. Cancelling 2^{a-1} gives $2b-1 = 2d-1$, hence $b = d$.
Surjective. Let $n \in \mathbb{N}$. Divide out factors of 2 until an odd number remains. Then uniquely

$$n = 2^k q, \qquad k \geq 0, \quad q \geq 1 \text{ odd}.$$

Every positive odd q has the unique form $q = 2b-1$ with $b = (q+1)/2 \in \mathbb{N}$. Put $a = k+1$. Then $f(a,b) = n$.
Therefore f is bijective. If $v_2(n)$ denotes the largest $k \geq 0$ for which $2^k \mid n$, then

$$f^{-1}(n) = \left(v_2(n) + 1, \ \frac{n/2^{v_2(n)} + 1}{2} \right).$$

For example, $40 = 2^3 \cdot 5$, so $f^{-1}(40) = (4, 3)$.

1.3 Question 3: when all permutations commute

Question

Let X be nonempty. Prove that $f \circ g = g \circ f$ for all bijections $f, g : X \to X$ if and only if X has at most two elements.

Approach and theory

Prove both directions. For $|X| \leq 2$, list all permutations. If X has at least three elements, construct two transpositions that do not commute.

Step-by-step solution

Suppose first $|X| = 1$. The identity is the only bijection, so all bijections commute. If $|X| = 2$, say $X = \{a, b\}$, there are exactly two bijections: id_X and the transposition τ interchanging a, b. The identity commutes with every map and $\tau\tau = \tau\tau$, so all pairs commute.
Conversely suppose X contains distinct a, b, c. Define f to interchange a, b and fix every other point; define g to interchange b, c and fix every other point. Then

$$(f \circ g)(a) = f(a) = b, \qquad (g \circ f)(a) = g(b) = c.$$

Since $b \neq c$, $f \circ g \neq g \circ f$. Hence the stated commutativity for all bijections forces $|X| \leq 2$.

1.4 Question 4: which relations are equivalence relations?

Question

Determine which of the following are equivalence relations.

(i) On $\mathbb{R}$: $a \sim b$ iff $a - b \in \mathbb{Q}$.

(ii) On $\mathbb{Q}$: $a \sim b$ iff $a - b \in \mathbb{Z}$.

(iii) On $\mathbb{Z}^*$: $a \sim b$ iff $b = 2^n a$ for some $n \in \mathbb{Z}$.

(iv) On $\mathbb{R}^2$: $(a, b) \sim (c, d)$ iff $a^2 + b^2 = c^2 + d^2$.

(v) On $\mathbb{N}$: $a \sim b$ iff some $n \in \mathbb{N}$, $n > 1$, divides both a and b.

(vi) On $\mathbb{R}$: $a \sim b$ iff $ab \in \mathbb{Q}$.

(vii) On $\mathbb{R}^*$: $a \sim b$ iff $0 < ab^{-1} \in \mathbb{Q}$.

Approach and theory

An equivalence relation must be reflexive, symmetric, and transitive. For a failure, one failed axiom and one explicit counterexample suffice. Relations defined by equality of an invariant are automatically equivalence relations.

Step-by-step solution

(i) **Yes.** Reflexive: $a - a = 0 \in \mathbb{Q}$. Symmetric: if $a - b \in \mathbb{Q}$, then $b - a = -(a - b) \in \mathbb{Q}$. Transitive: if $a - b, b - c \in \mathbb{Q}$, then $a - c = (a - b) + (b - c) \in \mathbb{Q}$.

(ii) **Yes.** The same proof works because $0 \in \mathbb{Z}$ and $\mathbb{Z}$ is closed under negatives and addition.

(iii) **Yes.** Reflexive: $a = 2^0 a$. Symmetric: $b = 2^n a$ implies $a = 2^{-n} b$. Transitive: $b = 2^n a$ and $c = 2^m b$ imply $c = 2^{m+n} a$. The equations themselves ensure the displayed quantities are integers in $\mathbb{Z}^*$.

(iv) **Yes.** Let $I(a, b) = a^2 + b^2$. The relation says $I(a, b) = I(c, d)$. Equality of real numbers is reflexive, symmetric, and transitive. Explicitly, equality reverses and chains exactly as required.

(v) **No.** It is not reflexive at 1. No integer $n > 1$ divides 1, so $1 \not\sim 1$. (It also fails transitivity: $2 \sim 6$ and $6 \sim 3$, but $2 \not\sim 3$.)

(vi) **No.** It is not reflexive for every real number. Take $a = \sqrt{\pi}$. Then $a^2 = \pi \notin \mathbb{Q}$, so $a \not\sim a$.

(vii) **Yes.** Here $a \sim b$ means a/b is a positive rational. Reflexive: $a/a = 1$. Symmetric: if $a/b = q > 0$ is rational, then $b/a = 1/q > 0$ is rational. Transitive: if $a/b = q > 0$ and $b/c = r > 0$ are rational, then $a/c = (a/b)(b/c) = qr > 0$ is rational.

Thus (i), (ii), (iii), (iv), and (vii) are equivalence relations; (v) and (vi) are not.

1.5 Question 5: equality of fractions

Question

Let $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ and define $(a, b) \sim (c, d)$ iff $ad = bc$. Prove this is an equivalence relation on X.

Approach and theory

The condition is equality of rational numbers $a/b = c/d$. We nevertheless verify all three axioms directly; in transitivity, nonzero denominators permit cancellation.

Step-by-step solution

Reflexive: $ab = ba$, so $(a, b) \sim (a, b)$.
Symmetric: if $ad = bc$, then $cb = da$, so $(c, d) \sim (a, b)$.
Transitive: suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then

$$ad = bc, \qquad cf = de.$$

Multiply the first equation by f: $adf = bcf$. Substitute $cf = de$ to get $adf = bde$. Since $d \neq 0$, cancel d in the integers to obtain $af = be$. Hence $(a, b) \sim (e, f)$. All three axioms hold.

1.6 Question 6: equality of differences

Question

Let $X = \mathbb{N} \times \mathbb{N}$ and define $(a, b) \sim (c, d)$ iff $a + d = b + c$. Prove this is an equivalence relation.

Approach and theory

Rearrange $a + d = b + c$ as $a - b = c - d$. The relation is equality of the integer-valued invariant $I(a, b) = a - b$.

Step-by-step solution

Reflexive: $a + b = b + a$. Symmetric: $a + d = b + c$ implies $c + b = d + a$, so $(c, d) \sim (a, b)$. For transitivity, assume $a + d = b + c$ and $c + f = d + e$. These say $a - b = c - d$ and $c - d = e - f$. Thus $a - b = e - f$, so $a + f = b + e$, hence $(a, b) \sim (e, f)$.

1.7 Question 7: classes in Questions 5 and 6

Question

Describe the equivalence classes for the relations in Questions 5 and 6.

Approach and theory

The class of an element consists of all elements with the same invariant: rational value a/b in Question 5 and integer difference $a - b$ in Question 6.

Step-by-step solution

For Question 5,

$$[(a, b)] = \{(c, d) \in \mathbb{Z} \times \mathbb{Z}^* : ad = bc\} = \{(c, d) : c/d = a/b\}.$$

Thus classes are indexed by rational numbers. The class corresponding to $q \in \mathbb{Q}$ contains every numerator–denominator representation of q.
For Question 6,

$$[(a, b)] = \{(c, d) \in \mathbb{N}^2 : c - d = a - b\}.$$

Thus classes are indexed by $k \in \mathbb{Z}$:

$$C_k = \{(c, d) \in \mathbb{N}^2 : c - d = k\}.$$

Every $k \in \mathbb{Z}$ occurs: for $k \geq 0$ use $(k + 1, 1)$, and for $k < 0$ use $(1, 1 - k)$.

1.8 Question 8: all classes for two standard relations

Question

Find all equivalence classes of (i) $a \sim b$ on $\mathbb{R}$ iff $a - b \in \mathbb{Q}$; and (ii) $a \sim b$ on $\mathbb{Z}$ iff $a - b$ is a multiple of a fixed $n \in \mathbb{N}$.

Approach and theory

Solve the defining condition for b. In (ii), use the division algorithm to choose one representative from each class.

Step-by-step solution

(i) For $a \in \mathbb{R}$,

$$[a] = \{b \in \mathbb{R} : a - b \in \mathbb{Q}\} = a + \mathbb{Q} = \{a + q : q \in \mathbb{Q}\}.$$

The collection of all classes is $\{a + \mathbb{Q} : a \in \mathbb{R}\}$. Distinct cosets are disjoint, although there is no simple finite list of representatives.

(ii) For $a \in \mathbb{Z}$,

$$[a] = a + n\mathbb{Z} = \{a + nk : k \in \mathbb{Z}\}.$$

By the division algorithm, every a has a unique remainder $r \in \{0, 1, \ldots, n - 1\}$. Hence the distinct classes are exactly

$$[0], [1], \ldots, [n - 1].$$

1.9 Question 9: one injective and one surjective composite

Question

Let $f : X \to Y$ and $g : Y \to X$. Suppose $g \circ f$ is injective and $f \circ g$ is surjective. Prove that both $f \circ g$ and $g \circ f$ are bijections.

Approach and theory

First extract properties of f: an injective composite gf forces f injective, while a surjective composite fg forces f surjective. Once f is bijective, use its surjectivity/injectivity to prove g injective/surjective.

Step-by-step solution

If $f(x_1) = f(x_2)$, applying g gives $(gf)(x_1) = (gf)(x_2)$. Since gf is injective, $x_1 = x_2$. Thus f is injective.
Let $y \in Y$. Since $fg : Y \to Y$ is surjective, there is $y_0 \in Y$ with $f(g(y_0)) = y$. Hence y is in the image of f, so f is surjective. Therefore f is bijective.
To show g injective, suppose $g(y_1) = g(y_2)$. Since f is surjective, choose $x_1, x_2 \in X$ with $f(x_i) = y_i$. Then $(gf)(x_1) = g(y_1) = g(y_2) = (gf)(x_2)$. Injectivity of gf gives $x_1 = x_2$, hence $y_1 = f(x_1) = f(x_2) = y_2$.
To show g surjective, take $x \in X$. Surjectivity of fg gives $y \in Y$ such that $f(g(y)) = f(x)$. Since f is injective, $g(y) = x$. Thus g is bijective. A composition of bijections is bijective, so both fg and gf are bijections.

1.10 Question 10: characteristic functions

Question

Let X be any set and $2 = \{0, 1\}$. For $A \subseteq X$, define $\chi_A : X \to 2$ by $\chi_A(x) = 1$ if $x \in A$ and 0 otherwise. Prove that $\chi : \mathcal{P}(X) \to 2^X$, $A \mapsto \chi_A$, is a bijection.

Approach and theory

For injection, equal characteristic functions must have the same 1-set. For surjection, recover a subset from any function $u : X \to \{0, 1\}$ by taking $u^{-1}(\{1\})$.

Step-by-step solution

Suppose $\chi_A = \chi_B$. For any $x \in X$,

$$x \in A \iff \chi_A(x) = 1 \iff \chi_B(x) = 1 \iff x \in B.$$

Thus $A = B$, so χ is injective.
Let $u \in 2^X$ be arbitrary. Define $A = u^{-1}(\{1\}) = \{x : u(x) = 1\}$. If $x \in A$, then $u(x) = 1 = \chi_A(x)$. If $x \notin A$, then $u(x) \neq 1$; since its only possible values are $0, 1$, $u(x) = 0 = \chi_A(x)$. Hence $u = \chi_A$, proving surjectivity.
The inverse is explicit:

$$\chi^{-1}(u) = u^{-1}(\{1\}).$$

Part II

Assignment 2: Magmas, Monoids, and Groups

Chapter 2

Assignment 2 solutions

2.1 Question 1: associativity and commutativity

Question

Determine which operations are associative or commutative:

(a) on $\mathbb{Z}$, $a * b = (a+3)(b+2)$;

(b) on $\mathbb{R}$, $a * b = a + ab$;

(c) on $\mathcal{P}(X)$, $A * B = C_X(A) \cup C_X(B)$;

(d) on any set X, $a * b = a$.

Approach and theory

Commutativity compares $a * b$ and $b * a$. Associativity compares $(a * b) * c$ and $a * (b * c)$. A single counterexample disproves a universal law.

Step-by-step solution

(a) Closure holds on $\mathbb{Z}$. Now

$$a * b - b * a = (a+3)(b+2) - (b+3)(a+2) = b - a,$$

so the operation is not commutative (for example $0 * 1 = 9$ but $1 * 0 = 8$). It is not associative: $0 * 0 = 6$, so

$$(0 * 0) * 0 = 6 * 0 = 18, \qquad 0 * (0 * 0) = 0 * 6 = 24.$$

(b) It is not commutative because $a * b = a + ab$ and $b * a = b + ab$; for example $1 * 2 = 3$ while $2 * 1 = 4$. Also

$$(a * b) * c = a + ab + ac + abc, qquada * (b * c) = a + ab + abc.$$

They are not always equal (take $a = b = c = 1$), so it is not associative.

(c) Since union is commutative, $A * B = A^c \cup B^c = B^c \cup A^c = B * A$; hence it is commutative. If $X \neq \varnothing$, take $A = B = \varnothing$ and $C = X$. Then $A * B = X$,

$$(A * B) * C = X * X = \varnothing, \qquad A * (B * C) = \varnothing * X = X.$$

Thus it is not associative for nonempty X. If $X = \varnothing$, $\mathcal{P}(X)$ has one element and the operation is automatically associative.

(d) For all a, b, c, $(a * b) * c = a * c = a$ and $a * (b * c) = a * b = a$, so it is associative. It is commutative exactly when $a = b$ for all $a, b \in X$, that is, when X has at most one element. In general it is not commutative.

2.2 Question 2: a commutative nonassociative law

Question

Prove or disprove: every commutative law of composition on a two-element set is associative.

Approach and theory

The claim is universal, so one symmetric operation table that fails associativity disproves it.

Step-by-step solution

Let $S = \{0, 1\}$ and define

$*$	0	1
0	1	0
1	0	0

The table is symmetric across the diagonal, so the operation is commutative. But

$$(0 * 0) * 1 = 1 * 1 = 0, \qquad 0 * (0 * 1) = 0 * 0 = 1.$$

Therefore it is not associative, and the statement is false.

2.3 Question 3: counting laws of composition

Question

Determine the number of laws of composition and commutative laws of composition on an n-element set.

Approach and theory

A law is a function $S \times S \to S$. For a commutative law, the table entry at (b, a) is forced by the entry at (a, b), so count only the diagonal and one triangular half.

Step-by-step solution

$S \times S$ has n^2 ordered pairs. For each pair, its output can be any of n elements, independently. Thus there are

$$n^{n^2}$$

laws.

For a commutative law, the n diagonal cells are free. For every unordered pair of distinct elements there is one free value, and there are $\binom{n}{2}$ such pairs. Thus the number of free

entries is

$$n + \binom{n}{2} = \frac{n(n+1)}{2}.$$

Each has n choices, so there are

$$n^{n(n+1)/2}$$

commutative laws.

2.4 Question 4: adjoining an identity

Question

Let $(S, *)$ be an associative magma, let $e \notin S$, and put $S' = S \cup \{e\}$. Define $a + b = a * b$ when $a, b \in S$, $a + e = a$, and $e + b = b$. Prove $(S', +)$ is a monoid with identity e.

Approach and theory

Closure follows from the definition. The equations involving e give the identity. For associativity, either all three elements lie in S, or at least one equals e and can be deleted from either parenthesization.

Step-by-step solution

For $a, b \in S'$, the definition always produces an element of S', so $+$ is a law of composition. It gives $e + a = a$ and $a + e = a$ for every $a \in S'$, including $a = e$; hence e is a two-sided identity.
Let $a, b, c \in S'$. If all lie in S, then

$$(a + b) + c = (a * b) * c = a * (b * c) = a + (b + c)$$

by associativity of $*$. If at least one is e, remove that e using the identity rule. For example, if $b = e$, both sides are $(a + e) + c = a + c$ and $a + (e + c) = a + c$. The cases $a = e$ and $c = e$ are identical, and if several are e the same deletion works. Thus $+$ is associative. Therefore $(S', +)$ is a monoid with neutral element e.

2.5 Question 5: are these magmas groups?

Question

Decide whether (a) $(\mathbb{Z}, *)$ with $a * b = a + b + ab$, and (b) $(\mathbb{R}_{>0}, *)$ with $a * b = a/b$, are groups.

Approach and theory

A group needs associativity, a two-sided identity, and an inverse for every element. One missing property is enough to disqualify it.

Step-by-step solution

(a) Closure is clear. Direct expansion gives

$$(a * b) * c = a + b + c + ab + ac + bc + abc = a * (b * c),$$

so it is associative. Also $0 * a = a = a * 0$, so the identity is 0. An inverse x of a would satisfy

$$a * x = a + x + ax = 0, \quad \text{so} \quad x = -\frac{a}{a+1}$$

when $a \neq -1$. For $a = 1$, this would be $-1/2 \notin \mathbb{Z}$; for $a = -1$, the equation becomes $-1 = 0$ and has no solution. Hence not every integer has an inverse, so this is not a group.

(b) Closure holds, but associativity fails:

$$(a * b) * c = \frac{a}{bc}, \qquad a * (b * c) = \frac{ac}{b}.$$

For example, with $a = b = 1, c = 2$, these are $1/2$ and 2. Hence it is not a group. In fact 1 is only a right identity ($a * 1 = a$); no fixed left identity can satisfy $e/a = a$ for every a.

2.6 Question 6: roots of unity

Question

For a positive integer n, let $G = \{z \in \mathbb{C} : z^n = 1\}$. Prove G is a group under complex multiplication.

Approach and theory

Check the four group requirements. Associativity is inherited from $\mathbb{C}$. For an inverse, use $z^n = 1$ to obtain $z^{-1} = z^{n-1}$.

Step-by-step solution

If $z, w \in G$, then $(zw)^n = z^n w^n = 1$, so $zw \in G$ (closure). Complex multiplication is associative. Since $1^n = 1$, $1 \in G$ and is the identity. If $z \in G$, then $z \neq 0$, and

$$z\, z^{n-1} = z^n = 1 = z^{n-1} z.$$

Thus $z^{-1} = z^{n-1}$, and $(z^{n-1})^n = (z^n)^{n-1} = 1$, so the inverse lies in G. Therefore G is a group. It is abelian because complex multiplication is commutative.

2.7 Question 7: the pointwise function group

Question

Let $(G, *)$ be a group and X nonempty. For $G^X = \{f : X \to G\}$ define $(f * g)(x) = f(x) * g(x)$. Prove $(G^X, *)$ is a group.

Approach and theory

Every group axiom is checked at an arbitrary input x. Equality of functions means equality of their values for every x.

Step-by-step solution

If $f, g : X \to G$, then $f(x) * g(x) \in G$ for every x, so $f * g$ is a function $X \to G$ (closure). For $f, g, h \in G^X$ and $x \in X$,

$$((f * g) * h)(x) = (f(x) * g(x)) * h(x) = f(x) * (g(x) * h(x)) = (f * (g * h))(x).$$

Thus the functions are equal and the operation is associative.
Let $E : X \to G$ be the constant function $E(x) = e_G$. Then $(E * f)(x) = e_G * f(x) = f(x)$ and $(f * E)(x) = f(x)$, so E is the identity.
For $f \in G^X$, define $f^{-1} : X \to G$ by $f^{-1}(x) = f(x)^{-1}$. Then

$$(f * f^{-1})(x) = e_G = (f^{-1} * f)(x)$$

for every x. Hence every f has an inverse and G^X is a group.

Part III

Assignment 3: Groups, Morphisms, Products, and Free Groups

Chapter 3

Assignment 3 solutions

3.1 Question 1: the group $\mathbb{Z}/n\mathbb{Z}$

Question

For $n \geq 1$, let $a \sim b$ on $\mathbb{Z}$ iff $n \mid (a-b)$, and let $\mathbb{Z}/n\mathbb{Z}$ be the set of classes. Show $\mathbb{Z}/n\mathbb{Z} = \{[0], \ldots, [n-1]\}$. Show there is a unique law $+_n$ for which $\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is a morphism of additive magmas, and prove $(\mathbb{Z}/n\mathbb{Z}, +_n)$ is a group.

Approach and theory

The division algorithm finds canonical representatives. The morphism condition forces $[a] +_n [b] = [a+b]$. We must prove this formula is independent of representatives, then check the group axioms.

Step-by-step solution

For any $a \in \mathbb{Z}$, the division algorithm gives unique integers q, r with $a = qn + r$ and $0 \leq r < n$. Then $n \mid (a-r)$, so $[a] = [r]$. Thus every class is among $[0], \ldots, [n-1]$. If $0 \leq r, s < n$ and $[r] = [s]$, then $n \mid (r-s)$ while $|r-s| < n$; hence $r - s = 0$. The listed classes are distinct.
If π is a morphism, then necessarily

$$[a] +_n [b] = \pi(a) +_n \pi(b) = \pi(a+b) = [a+b].$$

This proves uniqueness of a possible law. Define it by this formula. To prove well-definedness, suppose $[a] = [a']$ and $[b] = [b']$. Then $n \mid (a-a')$ and $n \mid (b-b')$, so

$$n \mid (a-a') + (b-b') = (a+b) - (a'+b').$$

Hence $[a+b] = [a'+b']$.
Closure is built into the formula. Associativity follows from integer addition:

$$([a] +_n [b]) +_n [c] = [(a+b)+c] = [a+(b+c)] = [a] +_n ([b] +_n [c]).$$

$[0]$ is an identity because $[a] +_n [0] = [a]$. The inverse of $[a]$ is $[-a]$, since $[a] +_n [-a] = [0]$. Therefore it is a group. Finally, $\pi(a+b) = [a+b] = [a] +_n [b]$, so π is indeed a morphism.

3.2 Question 2: affine maps

Question

For $a \neq 0$, $b \in \mathbb{R}$, let $f_{a,b}(x) = ax + b$, and let $G = \{f_{a,b} : a \neq 0, b \in \mathbb{R}\}$. Prove $(G, \circ)$ is a group.

Approach and theory

Compute a composition and an inverse explicitly. Associativity is inherited from composition of functions.

Step-by-step solution

For $a, c \neq 0$,

$$(f_{a,b} \circ f_{c,d})(x) = a(cx + d) + b = (ac)x + (ad + b),$$

so $f_{a,b} \circ f_{c,d} = f_{ac,ad+b} \in G$ because $ac \neq 0$. Thus G is closed. Function composition is associative.
The identity map is $f_{1,0}$. Solving $y = ax + b$ for x gives $x = (y - b)/a$, so

$$f_{a,b}^{-1} = f_{1/a,-b/a} \in G.$$

Indeed both composites have parameters $(1, 0)$. Therefore $(G, \circ)$ is a group.

3.3 Question 3: the group $\mathbb{Q}/\mathbb{Z}$

Question

On $\mathbb{Q}$, define $r \sim s$ iff $r - s \in \mathbb{Z}$. Let $\mathbb{Q}/\mathbb{Z}$ be the quotient. Prove there is a unique law $+$ making $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ a morphism of additive magmas, and conclude that $(\mathbb{Q}/\mathbb{Z}, +)$ is a group.

Approach and theory

This repeats the logic of Question 1 with equivalence modulo integers rather than modulo n. The forced formula is $[r] + [s] = [r + s]$.

Step-by-step solution

The morphism condition forces $[r] + [s] = \pi(r + s) = [r + s]$, proving uniqueness. Define the operation so. If $r - r' \in \mathbb{Z}$ and $s - s' \in \mathbb{Z}$, then

$$(r + s) - (r' + s') = (r - r') + (s - s') \in \mathbb{Z},$$

so the formula is well-defined.
Associativity follows from addition in $\mathbb{Q}$. The identity is $[0] = \mathbb{Z}$, and the inverse of $[r]$ is $[-r]$. Closure is automatic. Hence $\mathbb{Q}/\mathbb{Z}$ is a group, and the defining formula shows π is a morphism.

3.4 Question 4: subgroup decisions

Question

Determine whether: (i) $\mathbb{R}^+$ in $(\mathbb{R}, +)$; (ii) $\mathbb{R}^+$ in $(\mathbb{R} \setminus \{0\}, \cdot)$; (iii) $\mathbb{Q}^+$ in $(\mathbb{R}^+, \cdot)$; (iv) $\pi\mathbb{Q}$ in $(\mathbb{R}, +)$, are subgroups.

Approach and theory

Use the one-step subgroup test: a nonempty subset H is a subgroup iff $ab^{-1} \in H$ for $a, b \in H$; additively, iff $a - b \in H$.

Step-by-step solution

(i) No. The additive identity 0 is not positive; also the inverse $-a$ of a positive number is not positive.

(ii) Yes. It is nonempty, and for $a, b > 0$, $ab^{-1} = a/b > 0$. The subgroup test applies.

(iii) Yes. It is nonempty, and the quotient of two positive rationals is a positive rational.

(iv) Yes. It contains $0 = \pi \cdot 0$. If $\pi q, \pi r \in \pi\mathbb{Q}$, then $\pi q - \pi r = \pi(q - r) \in \pi\mathbb{Q}$ because $q - r \in \mathbb{Q}$. Notice that the irrationality of π is not required for closure, though it makes the description nontrivial.

3.5 Question 5: generated cyclic subgroups

Question

List the elements of: (i) $\langle i \rangle$ in $(\mathbb{C}^*, \cdot)$; (ii) $\langle \sqrt{2} \rangle$ in $(\mathbb{R}^+, \cdot)$; (iii) $\langle \sqrt{2} \rangle$ in $(\mathbb{R}, +)$; (iv) $\langle -\sqrt{2} \rangle$ in $(\mathbb{C}^*, \cdot)$; (v) $\langle 12 \rangle$ in $(3\mathbb{Z}, +)$.

Approach and theory

In a multiplicative group, $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$. In an additive group, $\langle g \rangle = \{kg : k \in \mathbb{Z}\}$.

Step-by-step solution

(i) Powers repeat every four: $\langle i \rangle = \{1, i, -1, -i\}$.

(ii) $\langle \sqrt{2} \rangle = \{(\sqrt{2})^k : k \in \mathbb{Z}\} = \{2^{k/2} : k \in \mathbb{Z}\}$.

(iii) $\langle \sqrt{2} \rangle = \{k\sqrt{2} : k \in \mathbb{Z}\}$.

(iv) $\langle -\sqrt{2} \rangle = \{(-\sqrt{2})^k : k \in \mathbb{Z}\}$. More explicitly it contains 2^m for even exponents $2m$ and $-2^m\sqrt{2}$ for odd exponents $2m + 1$, for every $m \in \mathbb{Z}$.

(v) $\langle 12 \rangle = \{12k : k \in \mathbb{Z}\} = 12\mathbb{Z}$, which is contained in $3\mathbb{Z}$.

3.6 Question 6: stabilizer and transporter

Question

Let $\mathrm{Sym}(X)$ be the group of bijections of nonempty X, and let $x \neq y$. Define $A_x = \{f : f(x) = x\}$ and $A_{x,y} = \{f : f(x) = y\}$. Prove A_x is a subgroup and $A_{x,y}$ is not.

Approach and theory

Use the subgroup test for A_x. For $A_{x,y}$, check the identity first.

Step-by-step solution

A_x is nonempty because $\mathrm{id}_X(x) = x$. Let $f, g \in A_x$. Since $g(x) = x$, applying g^{-1} gives $g^{-1}(x) = x$. Therefore

$$(f \circ g^{-1})(x) = f(g^{-1}(x)) = f(x) = x.$$

Thus $f \circ g^{-1} \in A_x$, so the subgroup test proves $A_x \leq \mathrm{Sym}(X)$.
But $\mathrm{id}_X(x) = x \neq y$, so $\mathrm{id}_X \notin A_{x,y}$. Every subgroup must contain the ambient identity. Therefore $A_{x,y}$ is not a subgroup (whether or not it is empty).

3.7 Question 7: which maps are group morphisms?

Question

Determine which are group morphisms:

(i) $f : (\mathbb{Z}, +) \to (\mathbb{Z}, +)$, $f(a) = 2a$;

(ii) $f : (\mathbb{R}, +) \to (\mathbb{R}, +)$, $f(x) = \cos x$;

(iii) $f : (\mathcal{P}(X), \triangle) \to (\mathcal{P}(Y), \triangle)$, $f(A) = A \cap Y$, where $Y \subseteq X$;

(iv) $f(A) = C_Y(A)$ between the same groups (meaning $Y \setminus (A \cap Y)$);

(v) evaluation $f : (\mathbb{R}^X, +) \to (\mathbb{R}, +)$, $f(\alpha) = \alpha(x_0)$.

Approach and theory

A homomorphism must preserve the operation and identity. For symmetric difference, membership is parity: an element lies in $A \triangle B$ iff it lies in exactly one of A, B.

Step-by-step solution

(i) Yes: $f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b)$.

(ii) No. The identity in $(\mathbb{R}, +)$ is 0, but $f(0) = 1 \neq 0$. Equivalently, $\cos(x + y)$ is not generally $\cos x + \cos y$.

(iii) Yes. For any $t \in Y$,

$$t \in (A \triangle B) \cap Y \iff t \text{ lies in exactly one of } A, B \iff t \in (A \cap Y) \triangle (B \cap Y).$$

Thus $f(A \triangle B) = f(A) \triangle f(B)$.

(iv) No (for nonempty Y). The identity is $\varnothing$, but $f(\varnothing) = Y \neq \varnothing$. If $Y = \varnothing$, both target and map are trivial and it is a homomorphism.

(v) Yes. Pointwise addition gives

$$f(\alpha + \beta) = (\alpha + \beta)(x_0) = \alpha(x_0) + \beta(x_0) = f(\alpha) + f(\beta).$$

3.8 Question 8: all homomorphisms $\mathbb{Z} \to \mathbb{R}$

Question

For $a \in \mathbb{R}$, define $f_a : \mathbb{Z} \to \mathbb{R}$ by $f_a(x) = ax$. Prove $f : (\mathbb{Z}, +) \to (\mathbb{R}, +)$ is a group morphism iff $f = f_a$ for some $a \in \mathbb{R}$.

Approach and theory

$\mathbb{Z}$ is generated additively by 1, so a homomorphism is determined by $f(1)$. Treat positive, zero, and negative integers.

Step-by-step solution

For every $a \in \mathbb{R}$, $f_a(m + n) = a(m + n) = am + an = f_a(m) + f_a(n)$, so f_a is a homomorphism.
Conversely, let f be a homomorphism and put $a = f(1)$. First $f(0) = 0$. For $n > 0$,

$$f(n) = f(\underbrace{1 + \cdots + 1}_{n \text{ times}}) = nf(1) = na.$$

Also $0 = f(n + (-n)) = f(n) + f(-n)$, so $f(-n) = -f(n) = -na$. Thus $f(k) = ak$ for every integer k, which says $f = f_a$.

3.9 Question 9: an injective cube homomorphism

Question

Let G be a group and $f : G \to G$ be $f(x) = x^3$. If f is an injective group morphism, prove G is commutative.

Approach and theory

The homomorphism property says $(xy)^3 = x^3y^3$. First derive that every square commutes with every cube. Injectivity then makes all squares commute; combining exponents 2 and 3 makes every square central. Return to the cube identity and cancel.

Step-by-step solution

Let $x, y \in G$. Since $yx = x^{-1}(xy)x$, conjugation gives

$$(yx)^3 = x^{-1}(xy)^3x.$$

Use the homomorphism property on both xy and yx:

$$y^3x^3 = (yx)^3 = x^{-1}(x^3y^3)x = x^2y^3x.$$

Cancel the rightmost x to obtain

$$y^3x^2 = x^2y^3. \tag{1}$$

Thus every square commutes with every cube.
In particular, $x^6 = (x^2)^3$ commutes with $y^6 = (y^3)^2$. Since $f(x^2) = x^6$ and $f(y^2) = y^6$,

$$f(x^2y^2) = x^6y^6 = y^6x^6 = f(y^2x^2).$$

Injectivity of f gives $x^2y^2 = y^2x^2$. Hence x^2 commutes with y^2. By (1), it also commutes with y^3. Therefore it commutes with $y = y^3(y^2)^{-1}$. Since x, y were arbitrary, every square is central.
Now the homomorphism identity and centrality of x^2, y^2 give

$$(xy)^2xy = (xy)^3 = x^3y^3 = x^2y^2xy.$$

Cancel xy on the right: $(xy)^2 = x^2y^2$. Thus $xyxy = xxyy$. Cancel x on the left to get $yxy = xyy$, then cancel y on the right to obtain $yx = xy$. Hence G is commutative.

3.10 Question 10: quotient of a product

Question

Let A, B carry equivalence relations $\sim_A, \sim_B$, and define $(a_1, b_1) \sim (a_2, b_2)$ iff $a_1 \sim_A a_2$ and $b_1 \sim_B b_2$.

(a) Construct the unique maps $(A \times B)/\sim \to A/\sim_A$ and $(A \times B)/\sim \to B/\sim_B$ making the projection squares commute.

(b) Prove these maps make $(A \times B)/\sim$ a product of the two quotients.

(c) Conclude the resulting map to $(A/\sim_A) \times (B/\sim_B)$ is a bijection.

Approach and theory

First factor each coordinate quotient map through the quotient of $A \times B$. For the universal property, a map into the quotient product must send t to the class of a pair of representatives; prove this is independent of those representatives. Finally write the canonical map explicitly and prove injection and surjection.

Step-by-step solution

Let $q : A \times B \to (A \times B)/\sim$ and let π_A, pi_B be the quotient maps.
(a) Define

$$r_A([(a, b)]) = [a]_A, qquadr_B([(a, b)]) = [b]_B.$$

If $[(a, b)] = [(a', b')]$, then $(a, b) \sim (a', b')$, so $a \sim_A a'$ and $b \sim_B b'$. Hence $[a]_A = [a']_A$ and $[b]_B = [b']_B$; both formulas are well-defined. They satisfy $r_Aq = \pi_Ap_1$ and $r_Bq = \pi_Bp_2$. They are unique because q is surjective: the commuting equation forces their value on every $[(a, b)]$.

(b) Let T be a set with maps $u : T \to A/\sim_A$ and $v : T \to B/\sim_B$. For each t, choose a_t, b_t with $u(t) = [a_t]_A$ and $v(t) = [b_t]_B$; quotient maps are surjective, so choices exist. Define

$$w(t) = [(a_t, b_t)].$$

If a'_t, b'_t are other choices, equality of classes gives $a_t \sim_A a'_t$ and $b_t \sim_B b'_t$, hence $(a_t, b_t) \sim (a'_t, b'_t)$. Thus w is well-defined. It satisfies $r_A w = u, r_B w = v$. If w' has the same coordinates, write $w'(t) = [(a', b')]$; equality of both coordinates implies $(a', b') \sim (a_t, b_t)$, so $w'(t) = w(t)$. Therefore w is unique. This is exactly the product universal property.

(c) The induced map is

$$\Phi([(a, b)]) = ([a]_A, [b]_B).$$

For any $([a]_A, [b]_B)$, $\Phi([(a, b)])$ equals it, so Φ is surjective. If $\Phi([(a, b)]) = \Phi([(a', b')])$, then $a \sim_A a'$ and $b \sim_B b'$, so $(a, b) \sim (a', b')$ and the source classes are equal. Thus Φ is injective and hence bijective.

3.11 Question 11: additive and multiplicative groups

Question

Prove $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \cdot)$ are isomorphic. Are $(\mathbb{Q}, +)$ and $(\mathbb{Q}^+, \cdot)$ isomorphic?

Approach and theory

Use the exponential map for real numbers. For rationals, use an isomorphism-invariant property: divisibility. In $(\mathbb{Q}, +)$ every element can be divided by every positive integer, but 2 has no rational square root multiplicatively.

Step-by-step solution

Define $E : \mathbb{R} \to \mathbb{R}^+$ by $E(x) = e^x$. Then $E(x + y) = e^{x+y} = e^x e^y = E(x)E(y)$, so it is a homomorphism. It is injective because the exponential is strictly increasing, and surjective because every $t > 0$ equals $e^{\log t}$. Hence it is an isomorphism, with inverse log. Suppose, for contradiction, that $\phi : (\mathbb{Q}, +) \to (\mathbb{Q}^+, \cdot)$ were an isomorphism. Choose $q \in \mathbb{Q}$ with $\phi(q) = 2$ (surjectivity). In $(\mathbb{Q}, +)$, $q = 2(q/2)$. Therefore

$$2 = \phi(q) = \phi(q/2 + q/2) = \phi(q/2)^2.$$

This says 2 is the square of a positive rational. If $r = a/b$ in lowest terms and $r^2 = 2$, then $a^2 = 2b^2$, so a is even; writing $a = 2c$ makes $b^2 = 2c^2$, so b is even, contradicting lowest terms. Thus no such isomorphism exists.

3.12 Question 12: rank of a finite free group

Question

Let $F(A), F(B)$ be free groups and suppose $F(A) \cong F(B)$. If A is finite, prove B is finite and $|A| = |B|$.

Approach and theory

Abelianization turns a free group on a set S into the free abelian group $\bigoplus_S \mathbb{Z}$. Isomorphisms preserve abelianization. Tensoring with $\mathbb{Q}$ turns free abelian groups into vector spaces whose dimension is the number of generators.

Step-by-step solution

For a group G, let $[G,G]$ be the subgroup generated by commutators $ghg^{-1}h^{-1}$. It is characteristic: any isomorphism $\phi : G \to H$ sends each commutator to a commutator and therefore induces

$$G/[G,G] \cong H/[H,H]. \tag{1}$$

We now identify the abelianization of $F(S)$. Send each free generator s to the basis vector e_s of $\bigoplus_{s\in S}\mathbb{Z}$. The universal property gives a surjective homomorphism. Its value on a word records the exponent sum of every generator. Its kernel consists precisely of words made trivial after imposing commutativity, which is $[F(S),F(S)]$. Hence

$$F(S)_{\mathrm{ab}} \cong \bigoplus_{s\in S} \mathbb{Z}. \tag{2}$$

Let $|A| = n$. From (1) and (2),

$$\mathbb{Z}^n \cong \bigoplus_{b\in B} \mathbb{Z}.$$

Tensor both sides over $\mathbb{Z}$ with $\mathbb{Q}$. An isomorphism remains an isomorphism, and we obtain

$$\mathbb{Q}^n \cong \bigoplus_{b\in B} \mathbb{Q}.$$

The vectors indexed by B form a basis of the right side, while the left side has dimension n. Isomorphic vector spaces have equal dimension. Therefore B has exactly n elements. In particular it is finite and $|B| = |A|$.

3.13 Question 13: a free rank-three subgroup

Question

Let $F = F(x,y)$. Prove that $u = x^2$, $v = y^2$, and $z = xy$ generate a subgroup isomorphic to the free group on u, v, z.

Approach and theory

Place the subgroup inside the index-two kernel of the parity map $F(x,y) \to \mathbb{Z}/2\mathbb{Z}$. A self-contained Schreier rewriting calculation produces a free basis x^2, yx^{-1}, xy. Replacing one basis element by its product with another is an invertible Nielsen move and changes yx^{-1} into y^2.

Step-by-step solution

Define $\epsilon : F(x,y) \to \mathbb{Z}/2\mathbb{Z}$ by $\epsilon(x) = \epsilon(y) = 1$; inverses also map to 1 because $-1 = 1$ modulo 2. Let $K = \ker \epsilon$. A word lies in K exactly when its total exponent sum modulo

2 is even. The two right cosets of K have representatives $1, x$.
We use the following rewriting fact, and prove the part needed here. If a free group $F(S)$ has a subgroup H with a set T of coset representatives containing every initial segment of each representative, define

$$\gamma(t,s) = ts\,\overline{ts}^{-1},$$

where $\overline{w} \in T$ represents the coset Hw. The nonidentity $\gamma(t,s)$ form a free basis for H. Indeed, insert $\overline{w_i}^{-1}\overline{w_i}$ between successive letters of a word $w = s_1 \cdots s_m$, where $w_i = s_1 \cdots s_i$. The factors telescope and rewrite every $h \in H$ as a product of the γ's. Conversely, expanding such a reduced product causes only the cancellations already encoded between consecutive coset representatives; a nonempty reduced product cannot become the empty word in the free group. Thus there is no nontrivial relation among the nonidentity γ's, proving both generation and freeness.
Apply this with $S = \{x, y\}$ and $T = \{1, x\}$. Computing the nontrivial Schreier generators (inverse letters supply the inverses of these) gives

$$\gamma(1,y) = yx^{-1}, \qquad \gamma(x,x) = x^2, \qquad \gamma(x,y) = xy.$$

Hence K is free on x^2, yx^{-1}, xy.
Now perform the basis change

$$yx^{-1} \longmapsto (yx^{-1})(xy) = y^2,$$

while fixing x^2 and xy. This change is invertible because $yx^{-1} = y^2(xy)^{-1}$. More abstractly, on a free basis (a, b, c) the assignment $(a, b, c) \mapsto (a, bc, c)$ has inverse $(a, b, c) \mapsto (a, bc^{-1}, c)$, so it preserves freeness. Consequently

$$x^2,\ y^2,\ xy$$

is a free basis of K. Therefore the subgroup generated by u, v, z is K and is isomorphic to the free group $F(u, v, z)$.

3.14 Question 14: eliminating a generator

Question

Prove that the group generated by x, y, z with the single relation $yxyz^{-2} = e$ is a free group.

Approach and theory

The relation contains x exactly once, so solve for x and eliminate it. To make this rigorous, construct inverse homomorphisms between the presented group and the free group on y, z.

Step-by-step solution

Let

$$G = \langle x, y, z \mid yxyz^{-2} = e \rangle.$$

The relation is equivalent to

$$yxy = z^2, \qquad \text{hence} \qquad x = y^{-1}z^2y^{-1}. \tag{1}$$

Thus x is completely determined by y, z.
Let $F(y,z)$ be the free group on y, z. Define

$$\alpha : F(y,z) \to G, qquad\alpha(y) = y, \quad \alpha(z) = z.$$

This is a homomorphism by the defining property of a free group. It is surjective because (1) expresses the remaining generator x using y, z.
Define a map on the generators of G by

$$\beta(y) = y, qquad\beta(z) = z, qquad\beta(x) = y^{-1}z^2y^{-1}quad\text{in } F(y,z).$$

Check the relator:

$$\beta(yxyz^{-2}) = y(y^{-1}z^2y^{-1})yz^{-2} = z^2z^{-2} = e.$$

Therefore the assignment respects the defining relation and gives a homomorphism $\beta : G \to F(y,z)$.
$\beta\alpha$ fixes y, z, so it is the identity on $F(y,z)$. The composite $\alpha\beta$ fixes y, z, and on x it gives $y^{-1}z^2y^{-1} = x$ by (1); hence it is the identity on G. Thus α, β are inverse isomorphisms and

$$G \cong F(y,z),$$

a free group of rank 2.

Final proof checklist

Before considering a solution complete, verify the following.

- For injectivity, did you begin with equal outputs? For surjectivity, did you construct a preimage of an arbitrary target?
- For an equivalence relation, did you separately prove reflexive, symmetric, and transitive?
- For a formula on classes, did you replace every representative and prove the value unchanged?
- For a group, did you check closure, associativity, a two-sided identity, and a two-sided inverse for every element?
- For a subgroup, did you check nonemptiness and ab^{-1} (or $a - b$)?
- For a homomorphism, did you use the correct operation in each group?
- For a free-group claim, did you prove there is no hidden relation, rather than merely failing to find one?